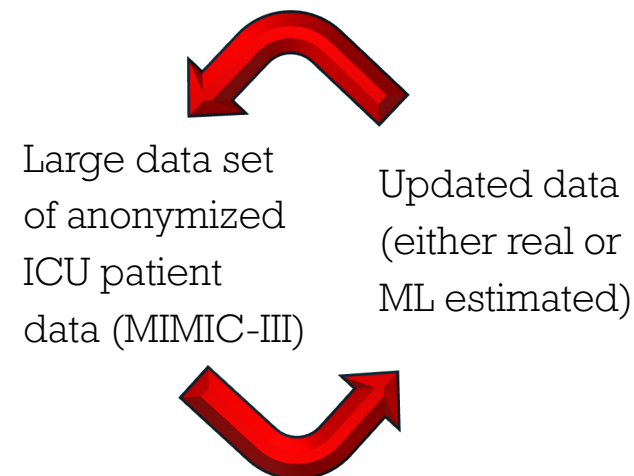


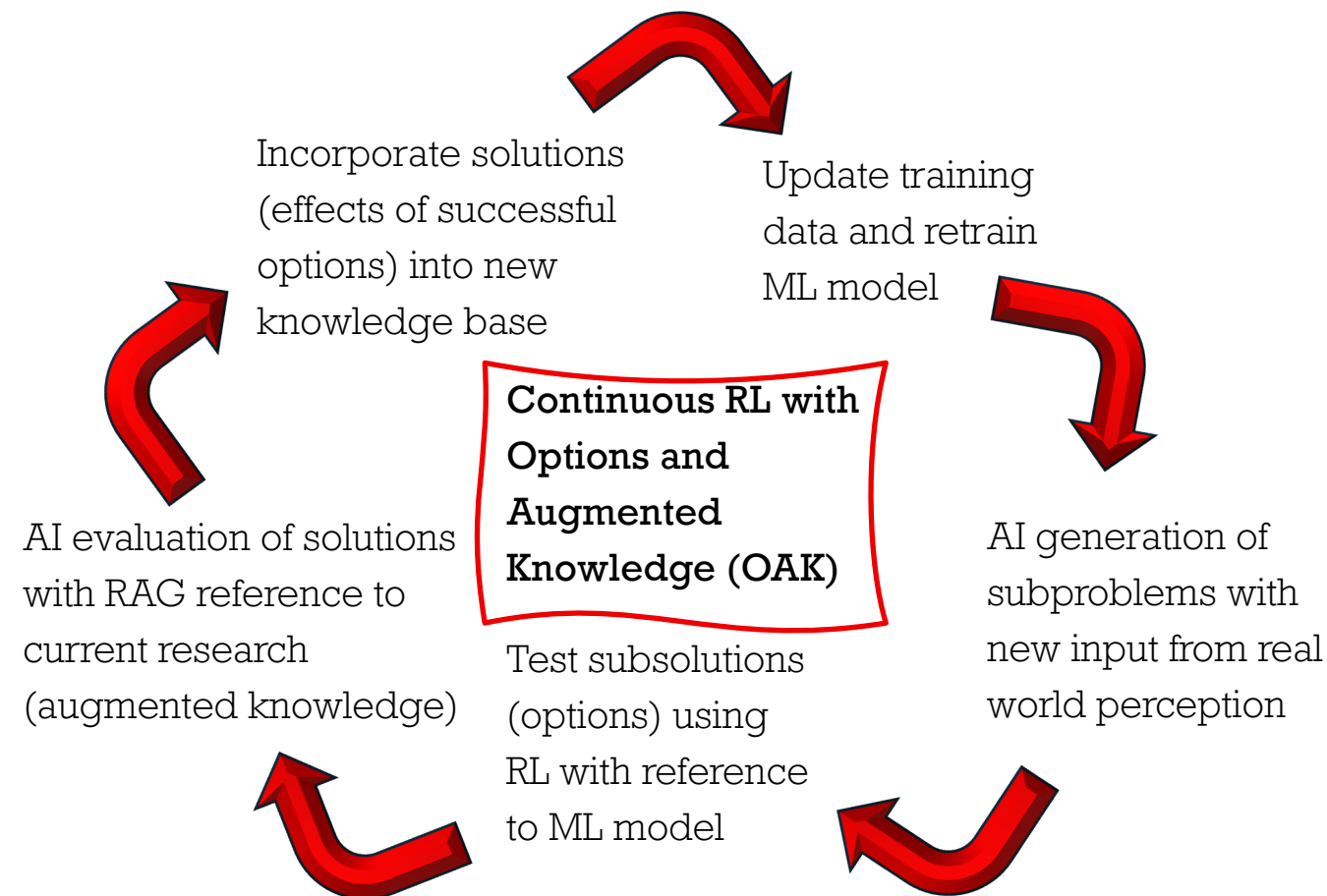
Quantum ML

As a proof-of-concept for quantum ML, we use a hybrid approach. Using IBM Qiskit tools, we train a quantum neural network (QNN), following Beer (2020), encoding input features from the MIMIC-III data set as qubits using RY gates, and a variational circuit of CZ gates and RY/RZ gates with learnable parameters. This allows a series of entangling and rotation operations to learn complex non-linear relationships in the data set. This is combined with Adaptive Moment Estimation (*Adam*) (Kingma, 2015) to adjust weights in a classical cross-entropy model for binary classification.



Towards a Framework for DDL

In a series of recent talks, Turing Award winner Prof. Richard Sutton has elaborated a speculative algorithm for Dynamic Deep Learning (2024). Sutton proposes the OaK (Options and Knowledge) approach in which ongoing background experimentation (options) gradually trains a continually updated ML model (knowledge). Sutton insists that such systems must be based on runtime continual learning rather than pre-trained models, while acknowledging that fundamental problems, namely the “loss of plasticity,” remain unresolved. We propose an intermediate experimental framework that leverages quantum computing to do runtime machine learning with access “augmented knowledge” from a corpus of human-composed content.



Augmented Knowledge

Continuing the proof-of-concept, we institute a simple iterative RAG pipeline to provide continually updated guidance for treating high-priority ICU patients identified hourly by the QML model. We use vector embeddings to match MIMIC-III cases to instances in the PMC-Patients dataset (Zhao, 2023) to fetch relevant medical research from PubMed. We generate treatment guidelines cross-checked against standard ICU practices. We propose a pipeline leveraging this augmented input in a simplified continual learning mechanism using a variation of Sutton's OaK algorithm.

